IoT-Based Real-Time College Bus Tracking and ETA Estimation System Using ESP32, GPS and MQTT

CS201 Introduction to IoT · Project Abstract (2026-27)

ABSTRACT

The college bus is scheduled to reach a stop at about 8:10 AM, but it arrives anywhere between 8:05 and 8:25 AM. Students cannot see where the bus is, so they either wait at the stop for a long time or are unsure when to leave home. We propose a low-cost IoT system that tracks the bus and shows its live position and estimated time of arrival (ETA) on a web page that works on a phone.

A battery-powered unit on the bus holds an ESP32 microcontroller, a NEO-6M GPS receiver and a 0.96-inch OLED display. The ESP32 reads NMEA location data from the GPS over UART, discards invalid fixes, smooths the speed, and works out the next stop and the distance to it on the device itself (edge processing). Every five seconds it sends a small JSON message over Wi-Fi to a cloud MQTT broker, using a phone hotspot as the internet connection. Two MQTT features help here. A retained message lets a newly opened page see the last known position straight away, and a Last Will message tells the system when the device has gone offline. The OLED shows the GPS fix, satellite count, connection status and next stop on the bus itself.

The web page subscribes to the broker, draws the bus on an OpenStreetMap map, calculates the ETA to the stop the student picks, and labels the data as live, stale or offline.

GPS does not work indoors, so the device has a replay mode that runs a stored route. This lets us demonstrate the prototype in the lab. The prototype covers one bus and a few dummy stops and is meant to cost about ₹2,000. It lets us show sensing, embedded programming, wireless communication, MQTT publish/subscribe, edge versus cloud processing and failure handling.

KEYWORDS

Internet of Things, ESP32, GPS, MQTT, bus tracking, edge computing, ETA estimation.